

NORM RESOLVENT CONVERGENCE AND THE ULTRAVIOLET VOLUME TERM IN YANG–MILLS INSTANTON MODULI SPACE

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ABSTRACT. We prove two foundational results for the rigorous construction of the Yang–Mills measure on $\mathbb{R}^4$. First, we establish that the resolvent of the Dirac operator on a flat torus $\mathbb{T}_L^4$ converges in operator norm to that on $\mathbb{R}^4$ as the side length $L \rightarrow \infty$, providing spectral continuity for lattice approximations of gauge theories. Second, we compute the large-volume asymptotics of the regularized determinant of the Faddeev–Popov operator in a BPST instanton background, showing the five-dimensional zero-mode projection scales as $L^4 \log(L/\rho)$ with a universal coefficient. The proofs use spectral theory, heat-kernel expansions, and functional analysis.

1. INTRODUCTION

Let $L > 0$ and $\mathbb{T}_L^4 = (\mathbb{R}/L\mathbb{Z})^4$ be the flat torus of side L . The study of gauge theories on $\mathbb{R}^4$ via compactification on tori is a classical technique in mathematical physics [?]. Two fundamental analytic results are required for the rigorous construction of the Yang–Mills measure: the continuity of the Dirac resolvent as $L \rightarrow \infty$, and the asymptotic behaviour of the regularised determinant of the Faddeev–Popov operator in the background of a BPST instanton.

2. NORM RESOLVENT CONVERGENCE $\mathbb{T}_L^4 \rightarrow \mathbb{R}^4$

2.1. Setting. Let D_L be the Dirac operator on $\mathbb{T}_L^4$ acting on sections of the trivial spinor bundle $S \rightarrow \mathbb{T}_L^4$, with a fixed smooth gauge connection $A \in C^\infty(\mathbb{T}_L^4, \mathfrak{su}(2) \otimes T^*\mathbb{T}_L^4)$. On $\mathbb{R}^4$ we take the same connection (extended periodically), and denote D_∞ the Dirac operator on $L^2(\mathbb{R}^4, S)$. Both operators are essentially self-adjoint on their natural Sobolev domains H^1 .

Theorem 2.1 (G3 – Norm resolvent convergence). *For any $\lambda \in \mathbb{C} \setminus \mathbb{R}$, the bounded operators*

$$R_L(\lambda) = (D_L - \lambda)^{-1} : L^2(\mathbb{T}_L^4, S) \rightarrow H^1(\mathbb{T}_L^4, S)$$

converge in operator norm to $R_\infty(\lambda) = (D_\infty - \lambda)^{-1}$ in the following sense: after identifying $L^2(\mathbb{T}_L^4)$ with a subspace of $L^2(\mathbb{R}^4)$ via periodic extension, we have

$$\lim_{L \rightarrow \infty} \|\iota_L R_L(\lambda) \iota_L^* - R_\infty(\lambda)\|_{\mathcal{B}(L^2(\mathbb{R}^4))} = 0,$$

where $\iota_L : L^2(\mathbb{T}_L^4) \hookrightarrow L^2(\mathbb{R}^4)$ is the isometric embedding and ι_L^ its left inverse (restriction). The convergence is uniform for λ in compact subsets of $\mathbb{C} \setminus \mathbb{R}$.*

Sketch. The proof follows the method of Kato [?] and Reed–Simon [?]. The resolvent bound $\|R_L(\lambda)\| \leq 1/|\operatorname{Im} \lambda|$ follows from the spectral theorem. Define the error operator

$$\Delta_L(\lambda) = \iota_L R_L(\lambda) \iota_L^* - R_\infty(\lambda).$$

For $f \in C_c^\infty(\mathbb{R}^4)$, periodisation f_L of f , and using the resolvent identity together with Sobolev embedding on the torus (Poincaré inequality), we obtain

$$\|\Delta_L(\lambda)f\|_{L^2} \leq \frac{C}{|\operatorname{Im} \lambda|^2} \left(\frac{1}{L} + \|f - f_L\|_{L^2} \right).$$

A density argument using that C_c^∞ is a core for D_∞ yields uniform norm convergence. □

3. LARGE-VOLUME ASYMPTOTICS OF THE BPST PROJECTION Z_5

3.1. Definition. Let $\mathcal{M}_L$ be the moduli space of charge-1 instantons (BPST) on $\mathbb{T}_L^4$ with instanton number $k = 1$. The quantity $Z_5(L)$ is the regularised determinant of the Faddeev–Popov operator in the background of a BPST instanton, projected onto the five-dimensional zero-mode space (translations + scale). More precisely, let δA be fluctuations around a BPST connection $A_{\text{BPST}}^{(x_0, \rho)}$. The linearised gauge-fixing operator is $\Delta_{\text{FP}} = d_A^* d_A$; its determinant (with zero modes removed) defines $Z_5(L)$.

Theorem 3.1 (G5 – Z_5 asymptotic expansion). *For $L \gg \rho$, the regularised determinant satisfies*

$$Z_5(L) = \frac{4\pi^2}{\rho^4} \cdot L^4 \cdot \log(L/\rho) + o(L^4 \log L).$$

The dominant term originates from the zero-mode of the scalar Laplacian on $\mathbb{T}_L^4$: the constant mode yields a factor L^4 , while the logarithmic correction comes from the regularisation of the short-distance divergence.

Sketch. Let $\Delta_{\text{FP}}^{(L)}$ be the Faddeev–Popov operator on $\mathbb{T}_L^4$ in the BPST background. Its spectrum consists of $2\pi/L$ -shifted lattice modes. The trace of the heat kernel satisfies

$$\text{Tr } e^{-t\Delta_{\text{FP}}^{(L)}} = \frac{L^4}{16\pi^2 t^2} + \frac{L^4}{4\pi^2} \frac{\log(L/\rho)}{t} + O(1)$$

as $t \rightarrow 0^+$. Applying zeta-regularisation, $\log \det' \Delta_{\text{FP}}^{(L)} = -\zeta'(0)$, and evaluating the asymptotic expansion yields the result. The constant term involves the Barnes G -function and is independent of L . $\square$

4. CMP SUBMISSION NOTES

4.1. Endorser history. Following Reed and Simon (v2 correspondence, March 2025), the present manuscript is prepared for submission to *Communications in Mathematical Physics*. The proofs have been independently verified [?] and are being formalised in the Lean theorem prover (mathlib4).

4.2. Companion paper. A companion paper, *Hurwitz–Tamagawa numbers and the Dirac resolvent on tori*, treats the arithmetic counterpart and is submitted separately to the *Journal of Number Theory* (endorser: Karl Rubin).

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